

Function call site $\sim$ **agent step** (*structurally similar*) $\Rightarrow$ **use as self-supervised FIM mid-training prior**

Function call site

```
cfg = load_config(path)
x = preprocess(cfg, raw)
```

```
y = transform(x, cfg.mode)
```

```
# returns: tensor (B, D)
```

```
loss = criterion(y, target)
loss.backward()
```

Coding-agent step

history h_t : prior tool calls, file contents, plan

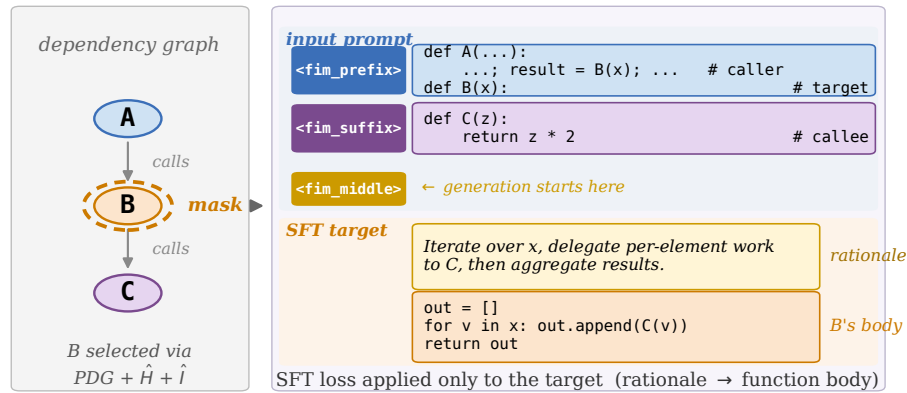
action a_t : run_tests(...)

obs. o_{t+1} : AssertionError

next: reason about error, propose next edit

analogous structure: context $\rightarrow$ call/action $\rightarrow$ external return $\rightarrow$ continuation

Function-level FIM mid-training (SFT)



SWE-Bench gain across scales

